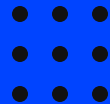
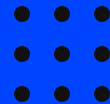
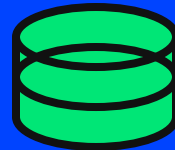
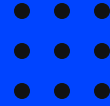


WHERE DOES AI GET ITS DATA?

FINDING, EVALUATING, AND BUILDING
DATASETS FOR MACHINE LEARNING

PRESENTER: GOKUL P



ROADMAP: TODAY'S ARC

8 WAYS TO GET DATA — RANKED EASIEST TO HARDEST

EASIEST TO ACQUIRE

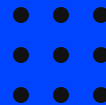
- S1** Ask an AI Model (S1)
Quick generation, synthetic seeds
- S2** Pre-Built Datasets — Kaggle & HF (S2)
Ready-to-use hosting hubs
- S3** UCI & Gov Open Data Portals (S3)
Structured public-domain stores
- S4** Augmentation & Synthetic Gen (S4)
Engineered programmatic extensions

Goal: Rapid prototypes & open academic benchmarks

HARDEST TO ACQUIRE

- S5** Closed/Restricted Research (S5)
Proprietary/siloed academic vaults
- S6** Reverse-Engineering Papers (S6)
Recreating published source methods
- S7** Mixing Multiple Datasets (S7)
Combining disparate public sources
- S8** Building Your Own From Scratch (S8)
Custom collection, labeling, & pipelines

Closing: Navigating common pitfalls & evaluation checklists



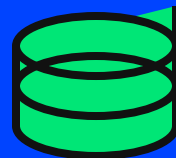
"Every model is a data problem before it's a model problem."

14M

ImageNet images

2.5 Yrs

to build



SOURCE 1: ASK AN AI MODEL

S1: EASIEST TO ACQUIRE

S1

WHAT IT IS

Prompt a model directly to create custom datasets

THE METHOD

"Generate me a synthetic CSV dataset for X with N rows..."

WHERE TO GO

claude.ai, chatgpt.com, gemini.google.com

STRATEGY & VELOCITY

BEST USED FOR

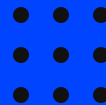
Prototyping downstream pipelines before production data is sourced

ENGINEERING VALUE

Perfect for verifying system architecture and testing code end-to-end

TIME TO FIRST DATASET

SECONDS



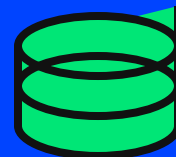
"Every model is a data problem before it's a model problem."

S1

Roadmap Rank

Instant

Generation



SOURCE 1: PROS & CONS

ADVANTAGES / PROS



Where AI Generation Excels

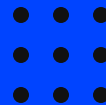
- **Instant & custom:** Zero setup needed to get a fully customized schema without licensing restrictions.
- **Stress-testing:** Ideal for testing and verifying engineering pipelines before production data becomes ready.

DISADVANTAGES / CONS



Inherent Structural Limits

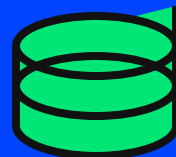
- **Artificial distribution:** The model does not understand the underlying true statistical distribution.
- **Hallucinated logic:** Can hallucinate highly implausible correlations that superficially look valid.
- **Not for performance:** Never usable as empirical evidence of real-world model accuracy.



"Every model is a data problem before it's a model problem."

S1

Synthetic Source



SOURCE 2: PRE-BUILT DATASETS

S2: ACQUISITION HUB

S2

THE CHANNELS

Kaggle & Hugging Face Datasets

THE METHOD

*Search/filter by task, size, or license →
kaggle datasets download or
load_dataset()*

WHERE TO GO

kaggle.com/datasets
huggingface.co/datasets

Instant setup via standard CLI/API tools

STRATEGY & UTILITY

BEST USED FOR

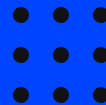
Standard, well-defined machine
learning problems and baseline models

COMMUNITY ADVANTAGE

Vetted licenses, open peer reviews,
and easily reproducible research
benchmarks

TIME TO FIRST DATASET

MINUTES



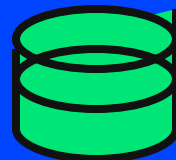
"Every model is a data
problem before it's a
model problem."

S2

Roadmap Rank

Minutes

Acquisition



SOURCE 2: PROS & CONS

ADVANTAGES / PROS



COMMUNITY BENEFIT

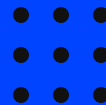
- **Huge selection:** Access to community-vetted materials, notebooks, discussions, and competitive leaderboards.
- **Instant ingestion:** One-line loaders that often ship with a baseline benchmark pre-configured.

DISADVANTAGES / CONS



INHERENT LIMITS

- **Highly overused:** Popular sets are heavily saturated; some are already baked inside pretrained models.
- **Variable quality:** Inconsistent quality controls and vague license clarity vary greatly set-to-set.



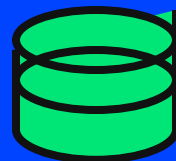
"Every model is a data problem before it's a model problem."

S2

Acquisition Hub

Public

Benchmarks



SOURCE 3: UCI & GOV PORTALS

S3: OPEN PUBLIC DOMAINS

S3

THE CHANNELS

UCI & Public Gov Repositories

THE METHOD

UCI for canonical small ML sets; gov portals for real-world civic, economic, or health data

WHERE TO GO

archive.ics.uci.edu

data.gov · data.gov.in · data.europa.eu

Classic sets or real-world civic domains

STRATEGY & UTILITY

BEST USED FOR

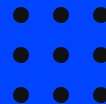
Learning classics (UCI) or grounded real-world domains (gov data)

DATA CHARACTERISTICS

Highly vetted statistical frameworks (UCI) paired with massive unvetted real-world civic tracking

TIME TO FIRST DATASET

MINUTES *



"Every model is a data problem before it's a model problem."

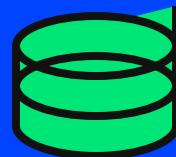
S3

Roadmap Rank

Minutes

Acquisition

* More browsing required to filter domains



SOURCE 3: PROS & CONS

ADVANTAGES / PROS



TRUST & DOCUMENTATION

Where Public Domains Excel

- **High trust:** High provenance and trust from institutional sources.
- **Clean & Real-world:** UCI sets are clean and well-documented, while government data realistically reflects real-world complexity and messiness.

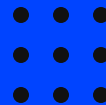
DISADVANTAGES / CONS



DATA & SYSTEM LIMITS

Inherent Pipeline Challenges

- **Aging benchmarks:** UCI datasets are often old, small, and highly over-benchmarked.
- **Gov data friction:** Public portals present inconsistent formats, missing metadata, and patchy overall documentation.



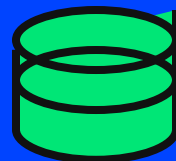
"Every model is a data problem before it's a model problem."

S3

Public Portals

Trusted

Institutional



SOURCE 4: AUGMENTATION

S4: EXPANSION ENGINE

S4

THE METHODOLOGY

Expanding Seed Datasets

- **What it is:** Expanding a small real seed set programmatically.
- **Efficiency:** Often achieved under 10 lines of code.
- **Pipeline:** Load seed data → apply transform pipeline → get N× more training samples.

Dynamic, programmatically scales seed data

STRATEGY & UTILITY

RELEVANT TOOLS

[Albumentations](#) / [imgaug](#)

(image flips, crops, noise)

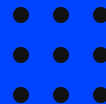
[SDV](#) / [CTGAN](#) (tabular) • [nlpaug](#) (text)

BEST USED FOR

Multiplying limited custom seed data into vast training distributions rapidly

TIME TO FIRST DATASET

MINUTES *



"Every model is a data problem before it's a model problem."

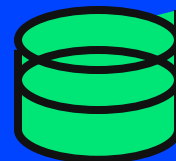
S4

Roadmap Rank

Minutes

Acquisition

* Scales infinitely once configured



SOURCE 4: PROS & CONS

ADVANTAGES / PROS



EFFICIENCY & TARGETING

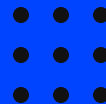
- **Cheap & Fast:** Genuinely a few lines of code; delivers the best effort-to-payoff ratio on this list.
- **Direct Targeting:** Can target specific gaps or boost rare classes directly.

DISADVANTAGES / CONS



BIAS & FIDELITY RISKS

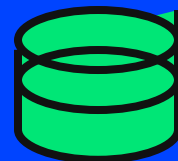
- **Bias Amplification:** Inherits and can amplify whatever bias is already in the seed data.
- **Extension, Not Replacement:** Never a substitute for real-world validation.



"Every model is a data problem before it's a model problem."

S4

Augmentation



SOURCE 5: RESTRICTED DATASETS

S5: SECURE CHANNELS

S5

THE PORTALS

Closed & Restricted Research

EXAMPLES & ACCESS METHOD

- **Credentialed access:** PhysioNet credentialed databases, hospitals, or specialized university repositories.
- **The workflow:** Apply for access sign a formal DUA verification of institutional affiliation & ethical approval.

Governed by strict institutional use policies

STRATEGY & UTILITY

BEST USED FOR

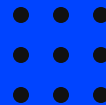
Highly sensitive, high-value domains like medicine, genetics, and finance with no public counterparts.

REGULATORY COMPLIANCE

Guarantees compliance with privacy metrics (HIPAA, GDPR) and formal clinical safety standards.

TIME TO FIRST DATASET

WEEKS *



"Every model is a data problem before it's a model problem."

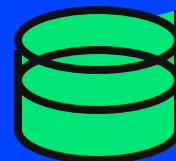
S5

Restricted

Weeks

Acquisition

* Requires vetting, ethics clearance, and signed DUAs



SOURCE 5: PROS & CONS

ADVANTAGES / PROS



RIGOR & EXCLUSIVITY

High Fidelity Benchmarks

Highest rigor: Real documentation and strict institutional oversight ensure maximum data quality.

Exclusive path: Often the single viable mechanism to leverage sensitive or high-value data sets.

DISADVANTAGES / CONS

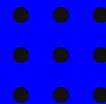


FRICTION & USAGE LIMITS

Inherent Regulatory Barriers

Slow acquisition: Exhaustive application processes and compliance checks delay baseline project launch.

Strict boundaries: Prohibits redistribution or publication of derived metrics, restricting usage strictly to approved purposes.



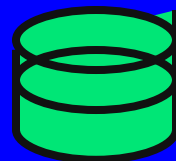
"Every model is a data problem before it's a model problem."

S5

Restricted Access

High

Quality Bar



SOURCE 6: REVERSE-ENGINEERING PAPERS

S6: DISCOVERY ENGINE

S6

THE METHODOLOGY

Mining Academic Pipelines

- **URLs:** paperswithcode.com · arxiv.org → then look for the paper's own GitHub/dataset page
- **How:** find the paper solving your problem → dig through its "Dataset" section or linked repository

Reconstruct benchmarks from academic source files

STRATEGY & UTILITY

BEST USED FOR

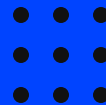
Tracking down custom novel pipelines, unique academic training sets, and state-of-the-art benchmarks directly.

WHY IT'S HARD

You're constructing someone else's pipeline from partial documentation, incomplete setup guides, and dead links.

TIME TO FIRST DATASET

Hours to Days*



"Every model is a data problem before it's a model problem."

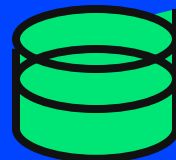
S6

Research Papers

Hours

Acquisition

* Requires extensive manual digging and custom pipeline rebuilding



SOURCE 6: PROS & CONS

ADVANTAGES / PROS



ACCURACY & BENCHMARKING

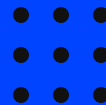
- **Research-Grade Fidelity:** Get exactly the dataset behind a specific published result.
- **Direct Comparison:** Lets you benchmark directly against the paper's own numbers.

DISADVANTAGES / CONS



DECAY & RECONSTRUCTION RISK

- **Access Barriers:** Links rot; downloads are often gated behind registration or access requests.
- **Detective Work:** Format is frequently undocumented outside the paper itself.



"Every model is a data problem before it's a model problem."

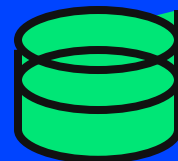
S6

Research Papers

High

Manual Effort

* Complex formatting requires manual pipeline reconstruction



SOURCE 7: MIXING MULTIPLE DATASETS

S7: DATASET INTEGRATION

S7

THE APPROACH

Blending Multi-Source Inputs

- **What it is:** Combining 2+ datasets on the same task from different sources for size/diversity.
- **How:** Harmonize schemas and label definitions across sources before merging.

STRATEGY & CHALLENGES

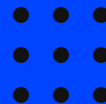
THE BARRIER

Why It's Hard

Every source brings its own custom format, unique labeling standard, and distinct license limitations.

TIME TO FIRST DATASET

Hours of Alignment*



"Every model is a data problem before it's a model problem."

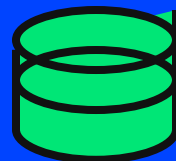
S7

Mixed Sources

Hours

Acquisition

** Requires tedious schema validation and metadata deduplication*



SOURCE 7: PROS & CONS

ADVANTAGES / PROS



SCALE & ROBUSTNESS

Synergistic Integration

- **Size & diversity:** Increases datasets in one move, covering critical gaps that no single source could satisfy.
- **Distribution testing:** Highly effective for establishing robustness checks across disparate data distributions.

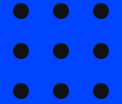
DISADVANTAGES / CONS



MISMATCHES & LICENSING

Inherent Integration Friction

- **Harmonization barriers:** Direct schema/label mismatches and uneven collection standards across sources.
- **Legal complexity:** Each source license must be vetted; a single restrictive term taints the entire merged dataset.



"Every model is a data problem before it's a model problem."

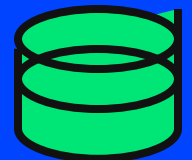
S7

Mixed Sources

High

Friction Bar

** Requires manual alignment of disparate schemas and individual licensing audits*



SOURCE 8: CUSTOM DATASETS

S8: CUSTOM GENERATION

S8

THE METHODOLOGY

Building From Scratch

- **What it is:** When nothing else fits — a novel problem, niche domain, or proprietary need.
- **Collection:** Scraping, sensors/hardware capture, crowdsourcing, or simulation (CARLA, Unity, Isaac Sim).

Simulation is key for robotics/vision where real physical capture is impractical.

PIPELINE & DEVELOPMENT

THE LIFECYCLE

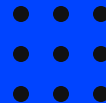
End-to-End Dataset Pipeline

**Source → Collect/Simulate → Label
→ Validate → Document → License**

Requires dedicated tooling like Label Studio, CVAT, or Roboflow to establish proper ground truth.

TIME TO FIRST DATASET

Days to Months*



"Every model is a data problem before it's a model problem."

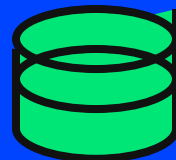
S8

Own Build

Max

Relative Cost

** The most expensive and labor-intensive data acquisition option on this list*



SOURCE 8: PROS & CONS

ADVANTAGES / PROS



CONTROL & FLEXIBILITY

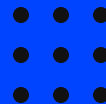
- **Perfect Fit:** Exact match for your problem — full control over schema, labels, and scope.
- **No Dependency:** No reliance on anyone else's data existing or being licensed right.
- **Simulation Power:** Generate scenarios too rare, dangerous, or expensive to capture for real.

DISADVANTAGES / CONS



OVERHEAD & VALIDATION RISK

- **Resource Intensive:** Extremely time- and resource-intensive.
- **No Benchmarks:** No existing baseline to check against — errors and bias can go unnoticed longer.
- **Total Burden:** You own 100% of labeling QC and the legal/ethics burden yourself.



"Every model is a data problem before it's a model problem."

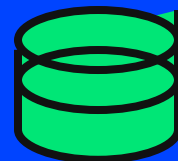
S8

Own Build

Max

Relative Cost

* High-investment custom generation requires dedicated pipeline oversight



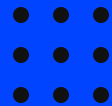
ACT 2: EVALUATING SOURCES

DATASET EVALUATION FRAMEWORK

Analyze, Prioritize, and Source

Evaluating the trade-offs of modern data acquisition pipelines:

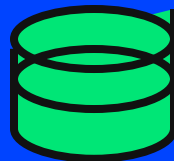
- Reviewing accuracy vs. operational overhead
- Balancing manual effort with licensing constraints
- Vetting custom pipelines for scale and robustness



"Every model is a data problem before it's a model problem."

ACT 2

Evaluate



EVALUATION: SIZE & COVERAGE

QUANTITY CRITERIA

Q1 CLASS DISTRIBUTION

Evaluating Class Gaps

Per-Class Sufficiency: Ensure you have enough examples per specific class, not just a high overall volume of data.

Distribution Gaps: A massive dataset with neglected rare classes fails to train robust generalized boundaries.

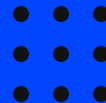
DENSITY & LIMITS

Q2 RANGE & TAILS

Distribution Coverage

Edge Cases: Audit whether your dataset actively covers rare edge cases or merely clusters around the "easy middle."

Hidden Caps: Undersized, biased, or imbalanced data quietly caps your model's maximum possible performance ceiling.



"Every model is a data problem before it's a model problem."

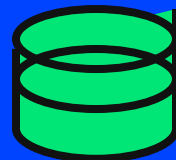
SIZE

And Coverage

Low

Ceiling Risk

* Insufficient tail representation quietly throttles final downstream model utility



EVALUATION: LABEL QUALITY

ANNOTATION ORIGIN

Q3

OWNERSHIP & PROVENANCE

Who Labeled It?

Labeling Source: Understand exactly who annotated your dataset and their baseline expertise level.

Disagreement Resolution: Ensure there was a clear, documented protocol for resolving conflicting annotations instead of ignoring them.

AGREEMENT METRICS

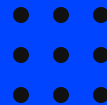
Q4

CONSISTENCY & ALIGNMENT

Inter-Annotator Agreement

Stated Scores: Having a documented consensus or agreement score is a strong signal of deliberate dataset hygiene.

The Red Flag: Having no stated alignment metrics is an immediate warning sign of highly divergent, low-confidence ground truth.



"Every model is a data problem before it's a model problem."

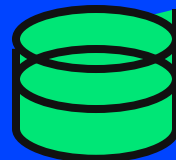
LABELS

Quality Control

High

Production Risk

** Noisy labels don't show up in metrics until it's too late — they manifest directly as silent failures in production.*



EVALUATION: BIAS & REPRESENTATION

DEMOGRAPHIC COVERAGE

Q5

REPRESENTATION AUDIT

Who Is Missing?

Imbalance Check: Understand exactly who or what is over-represented in the training data, and identify who is missing entirely from the sample set.

Real-World Gaps: A dataset that neglects entire demographics or geographic regions produces models that generalize poorly in multi-faceted environments.

SYSTEMIC BIAS

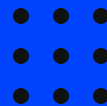
Q6

PREDICTIVE ALIGNMENT

Baked-In Bias

Data to Prediction: Bias baked into your training data inevitably becomes systemic bias baked directly into downstream model predictions and decisions.

The Gut-Check: Ask yourself, "If I only trained on this specific dataset, who would this system fail?" Identify the silent failures before deployment.



"Every model is a data problem before it's a model problem."

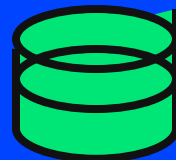
BIAS

Representation

Fatal

Systemic Risk

* Failing to represent the true distribution causes critical, unpredictable blindspots in production environments.



EVALUATION: LICENSE & CONSENT

TERMS OF ACQUISITION

Q7

INTENDED RE-USE

Authorized Scope

Release Validity: Was this data actually released for the use you have in mind? Vetting authorization boundaries is critical prior to downstream training.

Scraping Reality: "It's public" ≠ "it's licensed". Relying on scraped-without-permission data introduces substantial, active legal exposures.

PRIVACY AUDIT

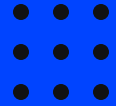
Q8

DATA PRIVACY

Consent Vetting

PII Red Flags: Strictly audit for names, faces, medical files, or financial records that persist without clear, documented explicit consent.

Anonymization Gaps: Even minor leakage of unmasked identifier fields creates compliance risk under global regulatory frame-works.



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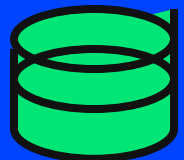
LEGAL

Consent & Rights

Risk

Legal Exposure

* Unauthorized data usage guarantees silent legal and regulatory liabilities in enterprise scale deployments.



EVALUATION: DOCUMENTATION

PROVENANCE RECORD

Q9

DATASET PROVENANCE

Collection Timeline

Acquisition Path: Do you know exactly how and when this dataset was collected and structured?

Traceability: Undocumented data leads directly to untraceable failures down the line where origin issues manifest unexpectedly.

METADATA CARDS

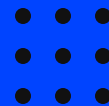
Q10

DOCUMENTATION STANDARDS

Model Cards & Specs

The Standard: Having no README or model card acts as a yellow flag rather than an automatic disqualifier.

Deep Verification: Always dig deeper and manually audit the sample metadata before placing complete trust in it.



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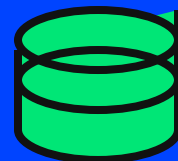
DATA

Provenance

Risk

Blind Failures

* Building models on unverified data guarantees untraceable anomalies in production systems.



EVALUATION: STANDARDIZATION

DATASHEET BLUEPRINT

Q11

DATASET SPECS

Gebru et al. (2018)

Hardware Analogy: Proposes that every dataset must be treated like a hardware component, shipping with a standardized datasheet.

Core Areas: Evaluates four essential pillars: motivation, composition, collection process, and intended use.

LIVING DOCUMENTATION

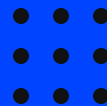
Q12

PORTAL INTEGRATION

HuggingFace Cards

Modern Standard: Dataset cards serve as the modern, living iteration of original hardware-inspired datasheets.

Hub Ecosystem: Directly filled in and maintained peer-to-peer on the Hub for every hosted dataset to enforce downstream transparency.



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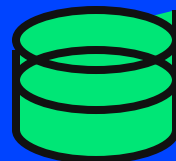
SHEETS

For Datasets

Audit

Transparency

** Without standardized datasheets, models inherit silent, untraceable data-level assumptions.*



EVALUATION: SOURCE LIMITATIONS

SYNTHETIC GENERATION

S1

ASK AN AI

Unrealistic Data

Generates highly structured yet fundamentally unrealistic distributions.

PUBLIC PORTALS

S2

KAGGLE & HF

Overused Hubs

Repetitive benchmarks with highly variable licensing clarity.

LEGACY REPOS

S3

UCI & GOVT

Outdated Formats

Often completely obsolete or structured inconsistently.

AUGMENTATION

S4

DATA SPINNING

Bias Amplified

Artificially multiplying samples amplifies baseline seed bias.

CLOSED RESEARCH

S5

PROPRIETARY LABS

Access Friction

Highly restricted usage rights coupled with intense access friction.

SCRAPED PATHS

S6

REVERSE ENGINEERING

Broken Links

Undocumented schemas combined with volatile, broken resource links.

HYBRID MIXING

S7

COALITION DATA

Schema Mismatch

Severe formatting mismatches paired with complex per-source license risk.

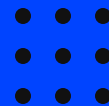
BESPOKE BUILDS

S8

BUILD YOUR OWN

Highest Effort

Massive resource sink with zero external benchmarks for validation.



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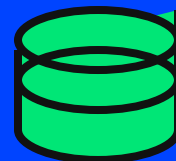
SOURCES

Limitations

Vetting

Bottlenecks

* Without thorough evaluation, models inherit invisible structural vulnerabilities from their origin pipeline.



EVALUATION: COMMON PITFALLS

INTEGRITY & LEAKAGE

P1

DATA CLEANLINESS

Contamination Risks

Train/Test Leakage: Information accidentally shared between splits invalidates performance validation and inflates metrics.

Near-Duplicate Samples: Redundant entries bias models and artificially blow up accuracy scores without generalizability.

ALIGNMENT & VIABILITY

P2

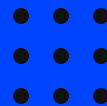
OPERATIONAL EXPOSURES

Distribution & Rights

Train-Serve Skew: Unrepresentative sampling creates critical performance drops in dynamic real-world pipelines.

Stale Data & Decay: The modern environment shifts while stagnant static datasets quickly lose semantic alignment.

License Infringements: Ignoring usage limitations introduces severe legal liabilities to operational systems.



"Every model is a data problem before it's a model problem."

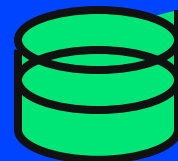
RISK

Data Quality

FAIL

Silent Drift

** Failing to anticipate structural leaks and target drift guarantees systemic failure under standard production conditions.*



TAKEAWAY CHEAT SHEET

ACQUISITION COMPLEXITY

E1

EASIEST → HARDEST

The Pipeline Spectrum

Synthetic AI: Fastest setup but prone to unrealistic bias.

Kaggle/HF Hubs: Centralized but variable licensing clarity.

UCI/Gov Repos: Stable origins but often outdated formats.

Augmentation: Easy multiplier that amplifies baseline seed error.

Closed Research: High friction access under restricted use.

Reverse-Engineered: High risk of broken resource links.

Mixed/Bespoke: Complex per-source licensing schema clashes.

Build-Your-Own: Highest effort with zero validation baseline.

VETTING FRAMEWORK

E2

EVALUATION METRICS

5-Point Safety Audit

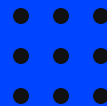
Dataset Size: Validate structural representation thresholds and balance before model design begins.

Explicit Labels: Audit for clean class alignments to prevent contamination and target leakage.

Systemic Bias: Identify and mitigate unrepresentative sampling and amplified seed bias.

Clear Licensing: Thoroughly vet terms of use to prevent silent operational liabilities.

Documentation: Require standard datasheets or model cards to track pipeline provenance.



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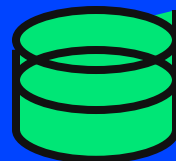
GUIDE

Quick Reference

SAVE

Screenshot Spec

** Always audit structural data limits and downstream compliance rights before starting training pipelines.*



THANK YOU!

CODE & WORKSPACE

R1

SOURCE DIRECTORY

Workspace & Files

Live Resources: All URLs shown live are compiled and collected in one clean repository place.

Slides & Demos: Access presentation slides plus interactive demo notebooks.

GitHub Path:

github.com/pgokul695/Where_does_A_I_gets_its_data

CONTACT & SOCIAL

R2

GET IN TOUCH

Connect Directly

Website: gokulp.in

Email: me@gokulp.in

LinkedIn: linkedin.com/in/pgokul695

GitHub: github.com/pgokul695

